Notes: Villalonga (JF, 2004)

Diversification Discount or Premium? New Evidence from the Business Information Tracking Series

Contents

1 Big picture

- **Question.** Is the well-known diversification discount a real valuation effect of corporate diversification, or is it partly an artifact of the COMPUSTAT segment data used to measure diversification?
- **Main contribution.** The paper uses the Business Information Tracking Series (BITS), an establishment-level Census database covering the U.S. private-sector economy, to construct firm business units more objectively and consistently than reported business segments.
- **Canonical benchmark.** The paper first verifies that, in the same BITS-COMPUSTAT Common Sample, COMPUSTAT segment data reproduce the familiar diversification discount.
- **Main reversal.** When the same firms are classified using BITS business units rather than COMPUSTAT segments, the discount becomes a diversification premium.
- **Core empirical result.** Table VI shows an average excess value of -0.18 for multisegment firms using COMPUSTAT segment data, but $+0.28$ for multibusiness firms using BITS data.
- **Interpretation.** The sign of the diversification effect depends on how the firm is divided into components and how comparable stand-alone business-unit values are constructed.
- **Why segment data are problematic.** Segment reporting is managerial, coarse, strategically discretionary, and subject to accounting rules that allow aggregation across activities. BITS records physical establishments and four-digit SIC industries, producing a more consistent classification.
- **Why BITS matters.** Unlike the Longitudinal Research Database, BITS covers the whole U.S. private-sector economy, not only manufacturing. This matters because most diversified firms operate partly or entirely outside manufacturing.
- **Takeaway.** Evidence about diversification discounts is highly sensitive to measurement. The paper challenges the view that diversification mechanically destroys value and shows that better business-unit data can overturn the conventional finding.

2 Incremental contribution of the paper

2.1 Contribution relative to the diversification-discount literature

- Earlier papers such as Lang and Stulz, Berger and Ofek, and Servaes use COMPUSTAT segments and find that diversified firms trade at a discount relative to focused firms.
- A second wave of work argues that the discount is partly due to selection: firms that diversify or firms acquired by diversifiers were already discounted before diversification.
- Villalonga's incremental contribution is different. She questions the *measurement foundation* of the discount itself.

Interpretation: The paper does not merely say the discount is endogenous. It shows that the discount can disappear and turn into a premium when diversification is measured with establishment-level business units instead of self-reported segments.

2.2 Contribution relative to segment-data critiques

- Segment data define units according to financial reporting rules and managerial discretion.
- Firms can strategically aggregate or split activities, and segment boundaries are not necessarily comparable across firms.

- BITS provides a common rule: all establishments with a common four-digit SIC code are aggregated into a business unit.

Interpretation: The paper contributes a cleaner measurement technology for decomposing diversified firms. This directly affects imputed q values and excess value estimates.

2.3 Contribution relative to establishment-level studies

- Maksimovic and Phillips and Schoar use plant-level manufacturing data to study diversification and productivity.
- Villalonga uses BITS, which covers nonmanufacturing establishments as well as manufacturing plants.
- This matters because a large share of diversified firms operate in services, retail, finance-related activities, or cross-sector portfolios.

Interpretation: The paper's full-economy coverage is an important advance. A manufacturing-only establishment database cannot fully measure diversification for conglomerates that operate outside manufacturing.

2.4 Contribution to empirical corporate finance methodology

- The paper shows that empirical conclusions can be reversed by changing the underlying unit of analysis.
- It emphasizes that industry benchmarks, imputed values, and diversification dummies are not innocuous constructed variables.
- It demonstrates that business-unit definitions and industry q measurement should be validated before drawing causal or normative conclusions from excess value regressions.

Interpretation: The broader lesson is methodological: corporate finance tests that rely on accounting-defined segments may confound economic diversification with reporting choices.

3 Data and measurement

3.1 Business Information Tracking Series

- BITS is a Census Bureau establishment-level database covering U.S. private-sector establishments with positive payroll over 1989–1996.
- An establishment is a single physical location where business is conducted or services are performed.
- BITS includes employment, payroll, primary four-digit SIC code, location, start year, firm identifiers, and legal entity identifiers.
- A BITS firm is the largest aggregation of business legal entities under common ownership and control.
- BITS covers all sectors except farms, railroads, the Postal Service, private households, and certain pension, health, and welfare funds.

Interpretation: BITS is valuable because it lets the researcher observe firms from the bottom up. Instead of trusting managers' segment reporting, the researcher observes where establishments operate and in which industries.

3.2 COMPUSTAT segment data

- COMPUSTAT segment files provide disaggregated financial data for segments that satisfy reporting materiality thresholds.
- A segment is an accounting unit representing a component engaged in providing products or services primarily to unaffiliated customers.
- Firms report segments at their discretion within accounting rules, so the same economic activities may be aggregated differently across firms or over time.
- Segment data include assets and sales, which are useful for value-weighting, but the segment boundaries may not coincide with economic business units.

Interpretation: The COMPUSTAT segment is a financial reporting object; the BITS business unit is an establishment-industry object. The paper asks whether that distinction changes valuation conclusions.

3.3 Common sample

- The main sample intersects BITS and COMPUSTAT.
- Firms must appear in both COMPUSTAT company and segment files and have market values available.
- Financial-sector firms are excluded.
- Agriculture, government, membership organizations, private households, and unclassified services are excluded from the establishment data.
- The BITS-COMPUSTAT Common Sample contains 12,708 firm-year observations from 1989–1996.

Interpretation: The common sample is designed so that BITS and COMPUSTAT can be compared on the same firms, avoiding the possibility that the reversal is driven simply by using different samples.

3.4 Business units versus segments

A BITS business unit is defined as all establishments of a firm with a common four-digit SIC code. A COMPUSTAT segment is the segment reported by the firm.

$$\text{Business unit}_{ij} = \{\text{establishments of firm } i \text{ with SIC code } j\}. \tag{1}$$

- BITS often classifies firms as more diversified than COMPUSTAT.
- Many firms that are single-segment in COMPUSTAT operate in multiple BITS business units.
- BITS detects within-segment diversification hidden by segment aggregation.

Interpretation: The BITS definition uses a common objective rule. This is why it often produces more business units than COMPUSTAT produces segments.

4 Excess value and industry q construction

4.1 Tobin's q

The paper uses a standard proxy for Tobin's q:

$$q_i = \frac{\text{market value of common equity}_i + \text{total assets}_i - \text{book value of common equity}_i}{\text{total assets}_i}. \tag{2}$$

Interpretation: This firm-level value is identical regardless of whether diversification is measured with BITS or segment data. The difference arises in the imputed stand-alone value.

4.2 Imputed q

For a firm with units j, imputed q is a weighted average of industry average q values:

$$\widehat{q}_i = \sum_j w_{ij}\bar{q}_{j,t}. \tag{3}$$

- With COMPUSTAT segments, weights are segment assets.
- With BITS business units, weights are business-unit employment.
- $\bar{q}_{j,t}$ is the average q of single-segment or single-business firms in the relevant industry-year.

Interpretation: The imputed value asks what the firm would be worth if its pieces were valued like comparable stand-alone firms. If the pieces are misdefined, the imputed value is mismeasured.

4.3 Excess value measures

The paper uses two excess value measures:

$$EV_i^{diff} = q_i - \widehat{q}_i, \tag{4}$$

$$EV_i^{ratio} = \log\left(\frac{q_i}{\widehat{q}_i}\right). \tag{5}$$

Interpretation: A negative excess value indicates a discount relative to imputed stand-alone value. A positive excess value indicates a premium.

4.4 Diversification measures

The main diversification variable is a dummy:

$$Diversified_i = 1\{\text{firm } i \text{ has more than one segment or business unit}\}. \tag{6}$$

The robustness tests also use continuous measures:

- number of business units;
- one minus Herfindahl index;
- total entropy;
- related entropy;
- unrelated entropy.

Interpretation: If the BITS premium is real rather than an artifact of a dummy definition, it should remain under continuous diversification measures. Table VIII shows that it does.

5 Empirical specifications

5.1 Mean excess value comparison

The simplest test compares diversified and focused firms:

$$\overline{EV}_{diversified} - \overline{EV}_{focused}. \tag{7}$$

Table VI reports this comparison for each year and pooled over all years using both COMPUSTAT segments and BITS business units.

Interpretation: This is the direct replication test. It asks whether the conventional segment-data discount appears in the common sample and what happens when BITS business units are used instead.

5.2 Pooled OLS robustness regression

The robustness regressions estimate

$$EV_{it} = \alpha + \beta Diversification_{it} + \Gamma' X_{it} + \varepsilon_{it}, \tag{8}$$

where X_{it} includes controls such as:

- log assets;
- EBIT-to-sales;
- capital expenditures-to-sales.

Interpretation: The coefficient β is the diversification premium or discount conditional on observed firm characteristics.

5.3 Industry q validation

Tables IV and V compare industry q measures across BITS and COMPUSTAT.

- BITS identifies more pure-play firms in many industry cells.
- COMPUSTAT single-segment firms often do not match BITS single-business industry classifications.
- BITS industry q values are closer to ideal measures in the subsample where an ideal benchmark can be formed.

Interpretation: This validation matters because excess value depends on the quality of industry benchmarks. If COMPUSTAT industry q is contaminated by hidden diversification, imputed values are biased.

6 Identification and interpretation

6.1 What the paper identifies

- The paper identifies how the measured diversification premium or discount changes when the unit of analysis changes from reported segments to objective business units.
- It does not randomly assign firms to diversify.
- It therefore does not prove that diversification causally increases firm value.
- It does show that the canonical diversification discount is not robust to better measurement of business-unit structure.

Interpretation: The paper's main target is a measurement claim: segment-based estimates are fragile. The premium result should not be read mechanically as proof that diversification always creates value.

6.2 Potential explanations for the reversal

The paper discusses two main explanations:

1. **Relatedness.** BITS may reveal that many diversified firms operate in related activities that are aggregated differently in COMPUSTAT.

2. **Strategic accounting.** Managers may choose segment definitions strategically, reporting segments in ways that affect measured diversification and benchmark values.

Interpretation: These explanations are not mutually exclusive. Both imply that segment data are a noisy and potentially biased measure of economic diversification.

6.3 Why all-sector coverage matters

- Manufacturing-only data are incomplete for studying conglomerates.
- Many diversified firms are entirely nonmanufacturing or are diversified across manufacturing and nonmanufacturing sectors.
- When the sample is restricted to pure manufacturing firms, the BITS premium becomes statistically insignificant.

Interpretation: The diversification question is an economy-wide corporate strategy question. Studying only manufacturing establishments can miss the activities that define many diversified firms.

7 Tables and figures

7.1 Tables I and II: sample structure and hidden diversification

Read:

- Table I counts firms, segments, business units, and establishments in the Common Sample and in COMPUSTAT from 1989–1996.
- BITS reports many more business units than COMPUSTAT reports segments.
- Table II shows that, in the Common Sample, 63.8% of firm-years are single-segment in COMPUSTAT, but only 20.6% are single-business in BITS.
- Many firms classified as focused using segment data are diversified using establishment-level business units.

Economic message: The conventional diversification measure misses a large amount of within-firm multi-industry activity.

Table I
Number of Firms, Segments, Business Units, and Establishments in the Common Sample, BITS, and COMPUSTAT

BITS refers to the *Business Information Tracking Series* of the U.S. Census Bureau. BITS covers all U.S. private-sector establishments with positive payroll in any year between 1989 and 1996, from both public and private firms. COMPUSTAT covers all firms that are publicly traded in U.S. stock markets. The figures reported for COMPUSTAT refer to all those active and inactive ("research") firms that appear in both the company and the segment COMPUSTAT files, and for which market values can be computed. The BITS–COMPUSTAT Common Sample has been constructed by merging both databases as described in the Appendix. An *establishment* is defined in BITS as "a single physical location where business is conducted or where services or industrial operations are performed," and a *firm* as "the largest aggregation of business legal entities (which are the legal owners of establishments) under common ownership and control." A *segment* is defined in SFAS 14 as "a component of an enterprise engaged in providing a product or service or a group of related products and services primarily to unaffiliated customers (i.e., customers outside the enterprise) for a profit." A *business unit* is defined in this paper as the aggregation of all of a firm's establishments with a common four-digit SIC code. Coverage ratios (i.e., the fraction of BITS or COMPUSTAT firms covered in the Common Sample) are in parentheses.

	Common Sample				BITS			COMPUSTAT	
Year	Establishments	Business Units	Segments	Firms	Establishments	Business Units	Firms	Segments	Firms
1989	192,908	13,227	2,680	1,481	6,063,857 (3.2%)	5,093,300 (0.26%)	4,978,250 (0.03%)	6,203 (43%)	3,929 (38%)
1990	197,606	13,167	2,787	1,563	6,126,016 (3.2%)	5,136,656 (0.26%)	5,024,252 (0.03%)	6,000 (46%)	3,754 (42%)
1991	211,603	13,688	3,011	1,762	6,155,181 (3.4%)	5,117,974 (0.27%)	5,005,347 (0.04%)	5,857 (51%)	3,687 (48%)
1992	214,973	14,269	3,269	1,999	6,275,349 (3.4%)	5,172,296 (0.28%)	5,051,405 (0.04%)	5,879 (56%)	3,745 (53%)
1993	197,253	12,559	2,801	1,700	6,356,799 (3.1%)	5,271,301 (0.24%)	5,149,208 (0.03%)	5,686 (49%)	3,665 (46%)
1994	183,626	11,729	2,558	1,549	6,465,057 (2.8%)	5,352,915 (0.22%)	5,232,956 (0.03%)	5,585 (46%)	3,649 (42%)
1995	177,937	11,006	2,340	1,403	6,566,634 (2.7%)	5,439,734 (0.20%)	5,322,981 (0.03%)	5,470 (43%)	3,620 (39%)
1996	179,465	10,641	2,121	1,251	6,699,635 (2.7%)	5,555,476 (0.19%)	5,439,206 (0.02%)	5,318 (40%)	3,528 (35%)
All firm-years	1,555,371	100,286	21,567	12,708	50,708,528 (3.1%)	42,139,652 (0.24%)	41,203,605 (0.03%)	45,998 (47%)	29,577 (43%)

Table I: sample counts

Table II
Number of Firm-Year Observations by Number of Segments or Business Units

The table reports the number (percentage) of firms in the sample that have one, two, or more segments (or business units). Segments and business units refer to the operations of a firm in a particular industry, according to COMPUSTAT and BITS, respectively. A *segment* is defined as "a component of an enterprise engaged in providing a product or service or a group of related products and services primarily to unaffiliated customers (i.e., customers outside the enterprise) for a profit." A *business unit* is the aggregation of all of a firm's establishments with a common four-digit SIC code. An establishment is defined in BITS as "a single physical location where business is conducted or where services or industrial operations are performed." The figures reported for COMPUSTAT refer to all those active and inactive ("research") firms that appear in both the company and the segment COMPUSTAT files, and for which market values can be computed. The Common Sample refers to the merger of the BITS and COMPUSTAT, as described in the Appendix.

Number of Segments or Business Units in Firm	Common Sample: Business Units, No. of Firm-Years	(Percentage)	Common Sample: Segments, No. of Firm-Years	(Percentage)	COMPUSTAT (Segments), No. of Firm-Years	(Percentage)
1	2,616	(20.6)	8,114	(63.8)	21,115	(71.4)
2	1,494	(11.8)	2,025	(15.9)	3,815	(12.9)
3	1,367	(10.8)	1,480	(11.6)	2,605	(8.8)
4	1,084	(8.5)	715	(5.6)	1,301	(4.4)
5	870	(6.8)	236	(1.9)	425	(1.4)
6	694	(5.5)	90	(0.7)	200	(0.7)
7	567	(4.5)	29	(0.2)	71	(0.2)
8	521	(4.1)	3	(0)	12	(0.04)
9	384	(3)	4	(0)	14	(0.05)
10	320	(2.5)	12	(0.1)	19	(0.06)
11–20	1,675	(13)				
21–30	636	(5.1)				
31+	480	(3.2)				
Total	12,708	(100)	12,708	(100)	29,577	(100)

Table II: firm-years by segments or business units

7.2 Tables III and IV: descriptive statistics and industry q construction

Read:

- Table III shows that Common Sample firms are broadly comparable to COMPUSTAT firms in employment, assets, and Tobin's q.
- It also shows major differences between multisegment and multibusiness classifications.
- Table IV documents differences between BITS and COMPUSTAT industry q values.
- BITS and COMPUSTAT use different pure-play firms to compute industry average q values.

Economic message: Industry benchmarks are central to the diversification discount. If industry q is mismeasured, excess value is mismeasured.

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Table III

Firm Employment, Assets, and Tobin's q: Means and Standard Deviations

Tobin's q is computed as the market value of common equity plus total assets minus the book value of common equity, divided by total assets. Data on both assets and q come from COMPUSTAT. Employment data come from both COMPUSTAT and BITS. COMPUSTAT employment figures represent the number of company workers as reported to shareholders, which for some firms is the average number of employees over the year and for others the number of employees at year-end. The figures include all employees of consolidated subsidiaries, both domestic and foreign, all part-time and seasonal employees, full-time equivalent employees, and company officers. BITS employment figures are for U.S.-based employees only and include full-time, part-time, and temporary employees, and salaried personnel. Employment in BITS is measured in the pay period that includes March 12 of every year. COMPUSTAT here refers to all active and inactive ("research") firms included in both the company and the segment COMPUSTAT files, and for which market values can be computed. The Common Sample refers to the merger of BITS and COMPUSTAT, as described in the Appendix. A *segment* is defined as "a component of an enterprise engaged in providing a product or service or a group of related products and services primarily to unaffiliated customers (i.e., customers outside the enterprise) for a profit." A *business unit* is the aggregation of all of a firm's establishments with a common four-digit SIC code. An establishment is defined in BITS as "a single physical location where business is conducted or where services or industrial operations are performed." The sample sizes on which these descriptive statistics are based are shown in Table I. Standard deviations are in parentheses.

	Common Sample				COMPUSTAT		
	Employment						
Year	BITS	COMPUSTAT	Assets ($Million)	q	Employment	Assets ($Million)	q
1989	8,454	10,157	1,748	1.7	7,145	1,468	2.0
	(27,324)	(31,045)	(8,590)	(1.8)	(25,551)	(6,955)	(4.5)
1990	8,040	9,594	1,766	1.6	7,347	1,647	1.9
	(26,749)	(29,659)	(8,963)	(1.8)	(25,668)	(7,706)	(13.2)
1991	7,360	8,901	1,672	1.9	7,663	1,792	2.2
	(24,500)	(28,480)	(8,792)	(2.3)	(25,973)	(8,214)	(9.5)
1992	6,630	8,059	1,601	1.9	7,651	1,893	2.2
	(23,212)	(26,778)	(9,233)	(1.7)	(25,741)	(9,004)	(10.2)
1993	7,200	8,560	1,797	1.9	7,914	2,220	2.1
	(24,707)	(27,412)	(10,331)	(1.6)	(26,220)	(10,526)	(5.6)
1994	7,651	9,197	1,928	1.7	8,263	2,498	1.8
	(24,981)	(28,647)	(9,866)	(1.1)	(27,496)	(11,607)	(2.3)
1995	8,367	10,049	2,187	1.9	8,735	2,878	2.2
	(26,349)	(30,929)	(11,308)	(1.5)	(28,358)	(13,457)	(5.9)
1996	9,492	11,896	2,668	1.9	9,524	3,378	2.3
	(28,446)	(34,848)	(13,170)	(1.4)	(30,106)	(15,725)	(11.5)
All firm-years	7,791	9,412	1,884	1.8	8,011	2,206	2.1
	(25,638)	(29,532)	(9,999)	(1.7)	(26,902)	(10,724)	(8.6)
Of Which:							
Multisegment	12,910	15,855	3,930	1.5	15,972	4,857	1.5
	(35,920)	(39,745)	(15,807)	(1.0)	(39,637)	(15,768)	(1.1)
Single-segment	4,893	5,765	725	2.0	4,721	1,143	2.3
	(16,609)	(20,854)	(3,379)	(2.0)	(18,346)	(7,584)	(10.1)
Multibusiness	9,726	11,706	2,342	1.6			
	(28,449)	(32,731)	(11,169)	(1.0)			
Single-business	326	563	119	2.6			
	(842)	(2,305)	(718)	(3.1)			
Single-segment but multibusiness	6,775	7,917	983	1.7			
	(19,424)	(24,417)	(3,963)	(1.1)			

Table III: firm employment, assets, and Tobin's q

Table IV

Industry Tobin's qs: Descriptive Statistics

Industry Tobin's q on BITS (COMPUSTAT) is computed as the average q of all single-business (single-segment) firms in the industry in any given year. Segments and business units refer to the operations of a firm in a particular industry according to COMPUSTAT and BITS, respectively, and are defined in the footnotes to Tables I and II. Averages are computed at the most precise SIC level for which there is a minimum of three single-business (single-segment) firms in the industry. Tobin's q is computed as the market value of common equity plus total assets minus the book value of common equity, divided by total assets. The sample is the BITS–COMPUSTAT Common Sample. The observations to which the percentages below refer are all the business unit-years or segment-years in the Common Sample (N = 100,286 for BITS-based industry qs; N = 21,567 for COMPUSTAT-based industry qs). Equivalently, the sample size is the number of different SIC codes in the sample weighted by the frequency with which each industry average is used in the computation of excess values. The Common Sample includes 889 different four-digit SIC codes in BITS and 655 in COMPUSTAT.

Panel A. Number of Pure-Play Firms in Industry Used to Compute Industry Average q

	3–5	6–10	11–20	21+	Total
BITS industry qs	30	16	14	40	100
COMPUSTAT industry qs	20	20	20	40	100

Panel B. Percentage of Industry qs, Business Units, and Segments, by SIC Level

	4-digit	3-digit	2-digit	1-digit	Total
BITS industry qs	13	13	40	34	100
COMPUSTAT industry qs	40	21	31	8	100
Business units defined at each SIC level in BITS	84	5	10	1	100
Segments defined at each SIC level in COMPUSTAT	98	2	0.3	0.2	100

Panel C. Summary Statistics for Industry qs

	Mean	*SD*	Within-Industry *SD*/Mean
BITS industry qs	2.48	1.13	0.25
COMPUSTAT industry qs	1.90	0.95	0.20

Table IV: industry Tobin's q statistics

7.3 Tables V and VI: industry q quality and main result

Read:

- Table V shows that COMPUSTAT segment SIC codes often fail to match BITS business-unit SIC codes.
- For single-segment but multibusiness firms, the reported segment SIC code often matches the largest business unit rather than the full set of business activities.
- Table VI reports the main reversal: segment data show a diversification discount, while BITS business-unit data show a diversification premium.
- In the pooled sample, the segment-data estimate is -0.18 and the BITS estimate is $+0.28$.

Economic message: The sign of the diversification effect flips when the unit of analysis is changed from reported segments to establishment-based business units.

Table V

Quality of the BITS and COMPUSTAT Industry *q*s

Industry Tobin's q on BITS (COMPUSTAT) is computed as the average q of all single-business (single-segment) firms in the industry in any given year. A *segment* is defined as "a component of an enterprise engaged in providing a product or service or a group of related products and services primarily to unaffiliated customers (i.e., customers outside the enterprise) for a profit." A *business unit* is the aggregation of all of a firm's establishments with a common four-digit SIC code. An establishment is defined in BITS as "a single physical location where business is conducted or where services or industrial operations are performed." Averages are computed at the most precise SIC level for which there is a minimum of three single-business (single-segment) firms in the industry. Tobin's q is computed as the market value of common equity plus total assets minus the book value of common equity, divided by total assets. The sample in Panel A is the subsample of 8,114 single-segment firms in the BITS–COMPUSTAT Common Sample. These include 2,616 single-business firms and 5,498 multibusiness firms. The sample in Panel B is the subsample of "ideal" industries, that is, all the four-digit industry codes for which there are three or more single-business firms. These firms are pure-plays according to both BITS and COMPUSTAT. The number of observations is 12,914 business units (or single-business firms) from the "ideal" industries. The asterisk indicates a 1 percent significant difference from the "ideal" q.

Panel A. Maximum Level of Precision at which BITS and COMPUSTAT SIC Codes Match

	4-digit	3-digit	2-digit	1-digit	None	Total
Single-segment, single-business firms	51	9	8	8	24	100
Single-segment but multibusiness firms:						
Segment SIC Code Matches Any Business Unit's SIC Code	84	6	4	3	3	100
Segment SIC Code Matches Largest Business Unit's SIC Code	14	1	2	2	81	100

Panel B. Comparison of Ideal *q*s and *q*s Computed at Different Aggregation Levels in BITS/COMPUSTAT

	4-digit	3-digit	2-digit	1-digit
BITS	*Ideal:* *2.56*	2.55	2.61*	2.57
COMPUSTAT	2.04*	2.04*	2.05*	1.94*

Table V: quality of BITS and COMPUSTAT industry q values

Diversified firms' excess value is computed as the mean difference in individual excess values between diversified (multisegment or multibusiness) firms and nondiversified (single-segment or single-business) firms. A *segment* is defined as "a component of an enterprise engaged in providing a product or service or a group of related products and services primarily to unaffiliated customers (i.e., customers outside the enterprise) for a profit." A *business unit* is the aggregation of all of a firm's establishments with a common four-digit SIC code. An establishment is defined in BITS as "a single physical location where business is conducted or where services or industrial operations are performed." Individual firms' excess values are computed as the difference between the firm's Tobin's q and its imputed q. Tobin's q is computed as the market value of common equity plus total assets minus the book value of common equity, divided by total assets. A firm's imputed q is the size-weighted average of the hypothetical qs of its segments (business units). Segment size is measured by assets; business unit size is measured by employment. A segment's (business unit's) hypothetical q is the average q of all single-segment (single-business) firms in its industry in any given year. Industry averages are computed at the four-digit SIC code level whenever possible. The sample is the BITS–COMPUSTAT Common Sample.

		Segment Data (COMPUSTAT)			Business Unit Data (BITS)		
Year	No. of Firms	Mean	(t-Stat)	% Firms Diversified	Mean	(t-Stat)	% Firms Diversified
1989	1,481	−0.29	(−4.09)	41	0.24	(2.11)	82
1990	1,563	−0.11	(−1.75)	40	0.40	(4.30)	80
1991	1,762	−0.21	(−2.63)	37	0.11	(0.77)	78
1992	1,999	−0.19	(−2.86)	34	0.29	(3.57)	74
1993	1,700	−0.18	(−2.48)	34	0.28	(2.90)	76
1994	1,549	−0.09	(−1.56)	15	0.43	(5.36)	80
1995	1,403	−0.22	(−2.59)	35	0.20	(1.63)	82
1996	1,251	−0.15	(−1.61)	37	0.35	(2.52)	86
All	12,708	−0.18	(−6.92)	36	0.28	(7.31)	79

Table VI: excess value in the common sample

7.4 Tables VII and VIII: robustness

Read:

- Table VII shows that the BITS premium is robust to several checks, including excluding firms with large non-U.S. activity and restricting to subsamples.
- The premium is weaker or insignificant for pure manufacturing firms, highlighting the importance of BITS all-sector coverage.
- Table VIII shows that the premium survives alternative excess value measures, alternative diversification measures, and control variables.
- Continuous measures such as entropy and one-minus-Herfindahl also produce positive diversification coefficients.

Economic message: The premium is not simply an artifact of a single dummy definition or a single excess value formula.

Table VII

Results of Robustness Checks

Diversified firms' excess value is computed as the mean difference in individual excess values between diversified (multisegment or multibusiness) firms and nondiversified (single-segment or single-business) firms, pooling all firm-year observations in the sample. A *segment* is defined as "a component of an enterprise engaged in providing a product or service or a group of related products and services primarily to unaffiliated customers (i.e., customers outside the enterprise) for a profit." A *business unit* is the aggregation of all of a firm's establishments with a common four-digit SIC code. An establishment is defined in BITS as "a single physical location where business is conducted or where services or industrial operations are performed." Individual firms' excess values are computed as the difference between the firm's Tobin's q and its imputed q. Tobin's q is computed as the market value of common equity plus total assets minus the book value of common equity, divided by total assets. A firm's imputed q is the size-weighted average of the hypothetical qs of its segments (business units). Segment size is measured by assets and business unit size is measured by employment, unless otherwise indicated. A segment's (business unit's) hypothetical q is the average q of all single-segment (single-business) firms in its industry in any given year. Industry averages are computed at the four-digit SIC code level whenever possible. The sample is the BITS–COMPUSTAT Common Sample. The number of observations is 12,708 unless otherwise indicated. The t-statistics are in parentheses.

	Segment Data			Business Unit data		
Robustness Check	Mean	(t-Stat)	% Firms Diversified	Mean	(t-Stat)	% Firms Diversified
Excess value estimates in Table IV	−0.18	(−6.92)	36	0.28	(7.31)	79
Variations in the computation of excess values						
Excluding firms with at least one business unit for which the industry q can only be computed at the one-digit level ($N = 5{,}705$)	−0.25	(−4.37)	36	0.18	(3.68)	79
Using segment employment as weights	−0.22	(−4.66)	36			
Using beginning-of-period employment as weights				0.13	(2.01)	79
Subsamples						
Firms with 100% operations in the United States ($N = 8{,}321$)	−0.21	(−6.48)	31	0.23	(5.11)	74
Pure manufacturing firms ($N = 2{,}332$)	−0.36	(−4.28)	19	0.03	(0.34)	49
Variations in the construction of business units						
Applying 10% materiality condition to business units				0.50	(18.24)	61
Including all vertically related activities in the same unit				0.31	(8.13)	75

In this sense, the premium found in BITS using employment weights can be considered a conservative estimate.

The use of employment as a measure of business unit size may also raise a

Table VII: robustness checks

Table VIII

Robustness of the Diversification Premium to Different Measures of Excess Value and Diversification, and to the Inclusion of Control Variables

This table reports the coefficients of the diversification variable from pooled OLS regressions of excess value on diversification. Multivariate regressions include three control variables: Log of assets; EBIT-to-sales; and capital expenditures-to sales. Individual firms' excess values are computed as the difference or the natural logarithm of the ratio of a firm's Tobin's q to its imputed q. Tobin's q is computed as the market value of common equity plus total assets minus the book value of common equity, divided by total assets. A firm's imputed q is the size-weighted average of the hypothetical qs of its segments (business units). Segment size is measured by assets; business unit size is measured by employment. A segment's (business unit's) hypothetical q is the average q of all single-segment (single-business) firms in its industry in any given year. Industry averages are computed at the four-digit-SIC code level whenever possible. The Herfindahl index of diversification is $\mathrm{H} = \sum_i P_i^2$, and the total entropy measure is $\mathrm{E_T} = \sum_i P_i^2 \ln(1/P_i)$, where P_i is the proportion of a firm's assets in industry i. Both measures are computed at the four-digit SIC level. Unrelated entropy, $\mathrm{E_U}$, is defined like $\mathrm{E_T}$ but computed at the two-digit SIC level. Related entropy is defined as $\mathrm{E_R} = \mathrm{E_T} - \mathrm{E_U}$. The sample is the BITS–COMPUSTAT Common Sample. The number of observations is 12,708 firm-year observations. The t-statistics are in parentheses.

	Measures of Diversification					
	Dummy	No. of Bus. Units	One Minus Herfindahl	Total Entropy	Related Entropy	Unrelated Entropy
Excess value measured as difference						
Univariate regressions	0.28	0.05	0.82	0.99	1.17	0.47
	(7.3)	(11.6)	(16.3)	(11.9)	(10.9)	(4.3)
Multivariate regressions	0.25	0.05	0.90	0.96	1.12	0.41
	(5.7)	(10.7)	(15.9)	(10.8)	(9.9)	(3.8)
Excess value measured as ratio						
Univariate regressions	0.47	0.06	0.97	1.23	1.35	0.69
	(29.8)	(35.6)	(53.1)	(37.6)	(31.2)	(15.0)
Multivariate regressions	0.38	0.02	0.96	1.08	1.13	0.58
	(21.0)	(26.5)	(45.8)	(31.1)	(24.6)	(13.1)

Table VIII: robustness to alternative measures and controls

7.5 Appendix Table AI: coverage of firms in the common sample

Read:

- Table AI examines how much of each firm's operations are covered in the BITS-COMPUSTAT match.
- Coverage ratios compare BITS employment to COMPUSTAT employment.
- The appendix shows that coverage ratios are not strongly correlated with the paper's key variables.

Economic message: The coverage analysis supports using the full matched sample rather than excluding firms with partial non-U.S. or unmatched operations.

Table AI

Coverage of Firms in the Common Sample

Coverage ratios (CR) for each firm are computed by dividing the firm's employment figure according to BITS, by its employment figure according to COMPUSTAT. COMPUSTAT employment figures represent the number of company workers as reported to shareholders, which for some firms is the average number of employees over the year and for others the number of employees at year-end. The figures include all employees of consolidated subsidiaries, both domestic and foreign, all part-time and seasonal employees, full-time equivalent employees, and company officers. BITS employment figures are for U.S.-based employees only and include full-time, part-time, and temporary employees, and salaried personnel. Employment in BITS is measured in the pay period that includes March 12 of every year. In this table, "industry" refers to the four-digit SIC code of the firm's largest segment in COMPUSTAT. The sample is the BITS–COMPUSTAT Common Sample. The asterisk denotes statistical significance at the 1 percent level.

Panel A. Coverage Ratios by Firm Type, Industry, and Industry q

	Employment		Coverage Ratio		No. of Observations	
	BITS (#)	COMPUSTAT (#)	Mean (%)	Median (%)	Total	w/CR > 100 (%)
All firm-years	7,791	9,410	123	90	12,708	32
By firm type:						
Diversified (multisegment)	12,910	15,855	111	90	4,594	30
Nondiversified (single-segment)	4,893	5,765	131	91	8,114	33
Diversified (multi-business)	9,726	11,706	132	91	10,092	32
Nondiversified (single-business)	326	563	90	89	2,616	33
Pure manufacturing	903	1,249	94	91	2,332	31
With nonmanufacturing operations	9,339	11,247	130	90	10,376	32
100 of operations in the US	5,664	5,502	131	97	8,321	42
Less than 100 of operations in the US	11,827	16,829	110	74	4,387	13
By industry sector						
Mining and construction	5,099	6,191	120	94	1,301	39
Manufacturing	8,732	11,642	129	89	3,079	28
Manufacturing	6,473	8,026	103	87	4,716	27
Transportation and communication	8,826	8,636	143	98	568	46
Wholesale and retail trade	15,460	16,383	127	96	1,472	39
Lodging and entertainment	4,749	6,234	184	90	10,938	33
Services	4,186	4,235	126	95	479	42
By tndustry q						
1st quintile ($q < 1.30$)	9,197	9,994	114	95	2,542	38
2nd quintile ($1.30 \leq q < 1.52$)	8,521	10,628	104	92	2,547	33
3rd quintile ($1.52 \leq q < 1.82$)	8,660	10,738	118	90	2,534	30
4th quintile ($1.92 \leq q < 2.36$)	7,879	9,705	138	89	2,542	30
5th quintile ($q \geq 2.36$)	4,701	6,000	143	85	2,543	29

Panel B. Correlations between Coverage Ratio and Key Variables

Employment					Excess Value	
BITS	COMPUSTAT	Assets	Firm q	Industry q	BITS	COMPUSTAT
0.079*	−0.026	−0.012	0.013	0.022	0.019	0.005

Table AI: coverage of firms in the Common Sample

8 Short summary

- The paper asks whether the diversification discount is partly a data artifact.
- COMPUSTAT segment data reproduce the discount in the BITS-COMPUSTAT Common Sample.
- BITS establishment-level business-unit data instead show a diversification premium.
- The key reason is that BITS defines business units objectively at the establishment-SIC level, while COMPUSTAT segments are self-reported and aggregated.
- BITS detects hidden diversification among firms that are single-segment in COMPUSTAT.
- BITS also provides better industry q benchmarks because pure-play firms are more consistently defined.
- Robustness checks show that the premium survives alternative diversification and excess value measures.
- The premium weakens for pure manufacturing firms, showing why all-sector establishment data matter.

- The paper's central message is that inference about diversification depends critically on measurement of business units and industry benchmarks.

9 Conclusion

9.1 Core conclusion

Villalonga shows that the diversification discount is not a robust fact once diversification is measured with establishment-level data. Segment data imply that diversified firms trade at a discount; BITS data imply that diversified firms trade at a premium.

9.2 Economic intuition

The diversification discount depends on comparing a firm's market value to the imputed value of its pieces. If the pieces are measured incorrectly, the discount is measured incorrectly. COMPUSTAT segments can hide business-unit diversification and contaminate industry benchmarks. BITS provides a more objective and consistent decomposition of firms into business units.

9.3 Takeaway

The paper does not prove that diversification always creates value. It shows that the canonical discount is fragile and may be a segment-data artifact. Researchers should be careful when interpreting excess value estimates based on managerial segment reporting.